IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

Stability and Control of Low-Carbon Electric Power Systems

Tuesday 23 May 10:00 am

Time allowed: 3:00 hours

Any special instructions for invigilators and information for candidates are on page 1.

ANSWER ALL QUESTIONS

Examiners responsible	First Marker(s) :	Dr Adria Junyent-Ferre; Dr Balarko Chaudhuri; Prof Tim Green
	Second Marker(s) :	

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Answer ALL Questions

Question 1 (30 marks)

a) Consider a three-phase inverter that uses SVPWM to generate the switching signals. The matrix of Clarke's transformation and a diagram of the 8 switching states of SVPWM are given in Figure 1.1 and equation (EQ1), respectively, for your reference.

$$C \triangleq \frac{2}{3}\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad \text{(EQ1)}$$

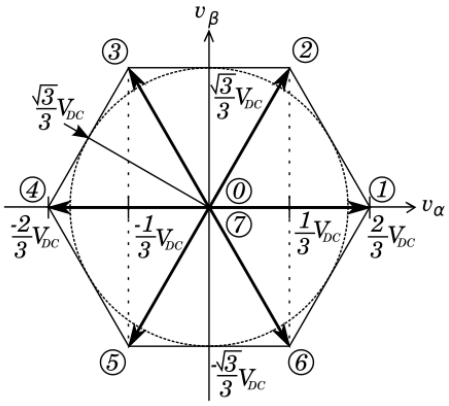


Figure 1.1: Switching vectors for SVPWM.

The inverter is connected in a three-wire configuration (ie without neutral conductor) and its DC-link voltage is 700 V. Consider the following three instantaneous voltage vectors and answer the questions below:

$$v_1^{\alpha\beta} = \begin{bmatrix} 361 \\ 97 \end{bmatrix} \text{V and } v_2^{abc} = \begin{bmatrix} 133 \\ 363 \\ -496 \end{bmatrix} \text{V}$$

i) Can the inverter apply these voltages? Justify your answer briefly.

[3]

ii) Should the controller limit the applied voltage using $\frac{\sqrt{3}}{3}V_{DC}$ instead of $\frac{2}{3}V_{DC}$? Justify your answer briefly.

[2]

iii) Switching states (eg ①, ②, ⑦…) would the inverter use to generate v_1?

[2]

iv) Calculate the times the converter would apply each switching state required to generate v_1. Consider a switching cycle of T_s= 100 μs (ie 10 kHz switching).

[5]

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b) Answer the following questions related to the steady-state analysis of a three-phase 10 kW inverter that is connected to a 50 Hz grid with an AC voltage of 400 V ph-ph rms. The DC-link voltage of the converter is 700 V and the converter uses an output filter based on a single reactor of X=0.25 pu.

i) Calculate the rated current of the converter (ie rated power delivered to the AC grid with unity power factor).

[3]

ii) Calculate the approximate voltage the bridge of the inverter would output if the output current was of 12 A aligned with the voltage of the grid.

[3]

iii) Explain briefly why the area of operation of the converter (ie the range of suitable P and Q of operation) is asymmetrical and the capacitive reactive power capability of the converter is lower than the inductive reactive power capability.

[3]

c) Provide brief answers (ie no more than 5-6 lines per question) to the following questions related to Phase Locked Loops (PLL) for inverter control:

i) What's the purpose of having a PLL in a Grid-Following inverter controller?

[3]

ii) Is it possible to make the tracking error of a PLL strictly zero at all times?

[3]

iii) What's the effect of having negative sequence in the voltage tracked by the PLL that is based on the Park transformation (ie a conventional three-phase PLL)?

[3]

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Question 2 (25 marks)

a) For a grid-forming inverter connected to an infinite bus, its behaviour is governed by the swing equation.

$$\frac{d^2\delta}{dt^2} = \frac{1}{J}\left(P'_{Mech} - K_D\frac{d\delta}{dt} - \frac{EV_{IB}}{X_S}\sin\delta\right)$$

Given the parameters below in per-unit values, answer the following questions.

$$P'_{Mech} = 1.0\text{pu}, E = V_{IB} = 1.0\text{pu}, X_S = 0.2\text{pu}$$

$$K_D = 1\text{pu}, J = 0.5\text{pu}$$

i) For the given parameters, calculate the angle δ at steady state.

[2]

ii) Find the linearised swing equation around the steady state.

[4]

iii) Find the undamped (by considering $K_D = 0$) natural frequency ω_n of the swing equation and describe the relationship between the inertia J and the natural frequency ω_n.

[4]

iv) Considering damping, find the poles of the linearised swing equation and describe how damping helps to stabilise the system.

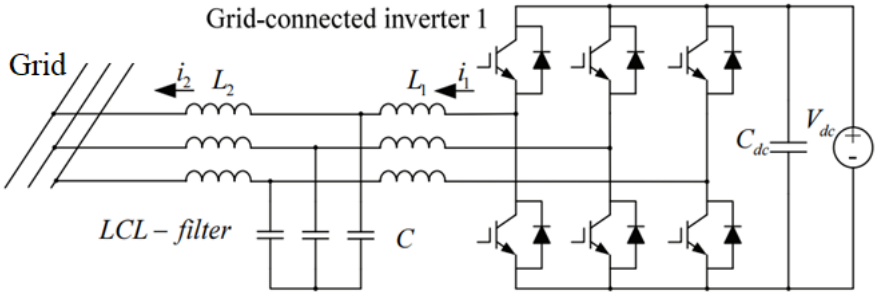
[4]

v) Does damping cause energy losses for grid-forming inverters? Explain the reason.

[2]

b) For a grid-connected inverter with an *LCL* filter shown in the figure below, the resonant frequency of the *LCL* filter is defined by:

$$f_{res} = \frac{1}{2\pi\sqrt{\frac{L_1 L_2}{L_1 + L_2} C}}$$



Given the parameters below, answer the following questions.

$$\text{Switching freqeuncy} = 10\text{kHz}$$

$$L_1 = 200\text{uH}, L_2 = 50\text{uH}, C = 160\text{uF}$$

i) The inverter control has a 1.5 switching-cycle time delay. Find the phase-shift caused by the time delay at the resonant frequency of the *LCL* filter and determine if the inverter is stable if a current control loop is used with a proportional-integral (PI) controller.

[4]

ii) Calculate the minimum value of capacitance required to ensure that the inverter is stable.

[2]

iii) Describe methods to be used to enhance the stability of the current loop.

[3]

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Question 3 (15 marks)

a) Bus incidence matrix is a way to capture the network topology to build a dynamic model of the system. Form the bus incidence matrix for the network in Figure 3 and show how it can be used to obtain the bus currents from individual branch/line currents and the difference between the voltages at the two ends of each branch/line. Use '+1' for the sending end (or from bus) based on the current directions shown on each line.

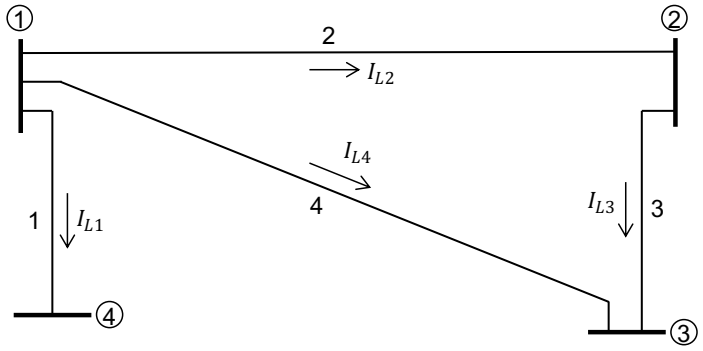


Figure 3: Single-line diagram of a network. Bus and line numbers and current directions are shown in the figure

[5]

b) Consider an aggregated model of an inverter-based resource (IBR) connected to a large (stiff) power system through two parallel transmission lines. Explain from a simple circuit point of view why the IBR could possibly destabilise when

i) it operates in grid-following mode with one transmission line out of service.

[3]

ii) it operates in grid-forming mode with both transmission lines in service.

[3]

iii) how can one deal with such instabilities in either of the above cases?

[1]

c) Explain why the q-axis component of the voltage control loop has no role to play in a simplified stability model of a grid-forming converter.

[3]

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